BGIN Layer2 Meetup @San Francisco

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12:30 - 13:00 Registration

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Layer2 meetup @ SF

13:00 - 13:15 Opening

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Layer2 meetup @ Tokyo

Blockchain Governance Initiative Network

Thank you for your sponsorship and support!

DGUS

Who we are

Blockchain Governance Initiative Network

BGIN is a non-profit organization that creates neutral forums for in-depth discussions among all stakeholders, documenting a common language to establish shared understanding. It also fosters innovation, best practices, guidelines, and standards for blockchain governance in operational, regulatory, and technical spheres.

Our History

Blockchain Governance Initiative Network

March 2019, Blockchain Roundtable
Regulators and academia from different countries, including Gary Gensler at MIT, discussed governance issues of blockchain.

June 2019, G20 Finance Ministers and Central Bank Governors Meeting (Fukuoka/Osaka, Japan)

G20 High-level Seminar on Financial Innovation

Key persons from different stakeholder groups discussed the possibility of multi-stakeholder governance for blockchain, which was reflected in the G20 Commuiqué.

Panelists:
Adam Back (CEO, Blockstream), **Brad Carr** (Senior Director, Digital Finance, International Institute of Finance), **Klaas Knot** (President, De Nederlandsche Bank, and Vice Chair, FSB), **Shin'ichiro Matsuo** (Research Professor, Georgetown University), **Jun Murai** (Professor, Keio University)

Sep 2019, FIN/SUM Japan

Key persons from different stakeholder groups discussed detailed issues and design of multi-stakeholder governance

March 2020, BG2C - Special Online Broadcasting Panel Discussion

Shin'ichiro Matsuo made public the launch of "Blockchain Governance Initiative Network, BGIN.

August 2020, BG2C. FIN/SUM Japan

Co-chairs of BGIN discussed BGIN's objectives and roadmaps

November 2020 - Present

BGIN hold 2-3 Block conferences per year in diverse regions of the world to discuss various blockchain governance topics

Block #1 (Mumbai/India), November, 2020
Block #2 (Paris/France) 8-10 March, 2021
Block #3 (DC/NY) 29 June - 1 July, 2021
Block #4 (Nairobi/Lome/Accra) 2-4 November, 2021
Block #5 (Virtual Japan/Tokyo) 4-6 April, 2022
Block #6 (Zurich/Switzerland), July 2022
Block #7 (Vancouver/Canada), December 2022
Block #8 (Bol/Croatia), May 2023
Block #9 (Australia), October 2023
Block #10 (Japan), March 2024
Block #11 (DC/US) October 2024 *Upcoming!

Blockchain Governance Initiative Network

Registration for BGIN Block#11 on Oct 21 and 22 in Washington, D.C is open! Click here!

Structure of Activities

Blockchain Governance Initiative Network

Layer 2 Meetups
(in-person, 1 or 2 times months)

Collocate with other events including Bitcoin events, ETH events, Fintech Weeks, TC307, Internet Identity Workshop (IIW), etc

- Discussion on project documents in progress with new community members
- Discovery of new topics for discussion at Block
- Networking

Block Conferences
(in-person, twice a year)

- Approval/multi-stakeholder discussion on project documents created or in progress
- Clarification of discussion points regarding new topics
- New projects launch
- Steering Committee Meeting

Working Groups
(remote, bi-weekly)

- Project progress management
- Discussion on the project details
- We currently have 1) Identity, Access, and Key Management and Privacy WG and 2) Financial Applications and Social Economics WG

What we create

BGIN has created documents that contribute to sound blockchain governance, attracting interest from various stakeholder groups and potentially leading to standards creation.

Blockchain Governance Initiative Network

Published

- Present and Future of a Decentralized Financial System and the Associated Regulatory Considerations
- Study Report for Ransomware Reaction
- NFT Study Report Part 1: Introduction and Use Cases]
- Incident Response of Decentralized Custody; A Case Study
- Soulbound Tokens (SBTs) Study Report Part 1: Building and Embracing a New Social Identity Layer?
- Proposal of Principles of DeFi Disclosure and Regulation
- Study Report on Selective Disclosure: Overview and Classifications
- Potential Points of Failure of Stablecoins and DeFi

Work in progress

- Soulbound Tokens (SBTs) Study Report Part 2
- Privacy vs Discoverability
- Taxonomy of Illicit Activities
- Layer 2 Governance
- Blockchain Analytics Research
- Harmonisation of different digital tokens in the future financial system
- The Mutual Impact and Improved Outcomes of Blockchain & AI

Link to the documents: https://bgin-global.org/documents

WORKSHOP RULE

Blockchain Governance Initiative Network

CHATHAM HOUSE RULE

When a meeting, or part thereof, is held under the Chatham House Rule, participants are free to use the information received, but neither the identity nor the affiliation of the speaker(s), nor that of any other participant, may be revealed.

All workshop sessions in Layer2 Meetup will be held in Chatham House Rule.

WORKSHOP RECORDING USAGE

The recording of all workshop sessions will not be distributed and will not be used internally to create anonymized reports, which will be published on BGIN's website.

Layer2 meetup @ SF

Blockchain Governance Initiative Network

13:15 - 14:45 Workshop Session 1

Layer 2 Governance

sponsored by DGUS

WORKSHOP RULE

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Introduction

- Layer 2 scaling solutions are being developed as a way to address the scalability and cost-efficiency challenges of Ethereum. But we also need to acknowledge that the trade-off for achieving higher scalability may introduce new security and governance challenges.
- At present, a research that objectively analyzes the trust points created by Layer 2 solutions can serve as a valuable resource for businesses and authorities to understand the technical limitations and possibilities and discuss how to solve them.
- Whether regulatory intervention is necessary to resolve the inevitable security and governance challenges, or whether there are appropriate technological solutions currently under development, is likely to become a topic of future discussion.
- Given the multitude of projects aiming to build blockchain-based financial systems, it is worth considering how contributions can be made to address the security and governance challenges, and their solutions, when using Layer 2 scaling solutions to enhance scalability.

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Definition: L2

Blockchain?

Yes
Blockchain-based L2

No
Other L2
e.g., State channel

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Definition: Governance

- The definition of a ‘governance system’ is all about changing the rules of the game.
- We can change the rules for L2s through a governance multi-sig or governance tokens
- ‘Governance’ can be sometimes seen as a magical tool to solve any problem; to fix bugs, to add features, to add incentives. etc
- Governance comes with one major downside: It can be an attacker vector.

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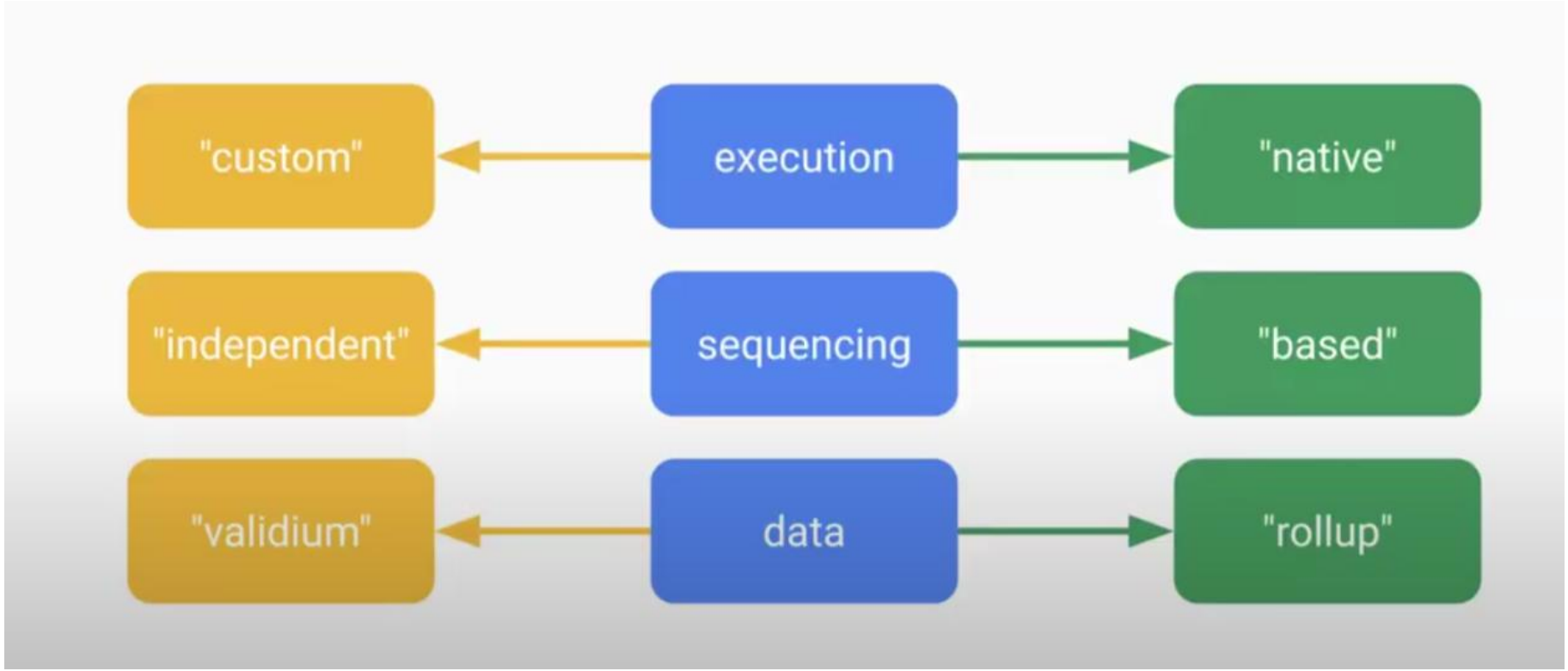
Recap of BGIN Block#10



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L1/L2 Operation

	State Channel	Plasma	Optimistic Rollup	ZK-Rollup	Validium
Use case	Quick micropayments, real-time transactions	Services that requires the efficiency for many micro payments	Services that need cost efficiency	Services that need relatively high security	Services that require relatively high throughput
Data	Off-chain (stored only in local environments)	Off-chain (data storage relies on operators but each user holds data by themselves)	On-chain (stored as calldata without compression)	On-chain (stored as calldata with compression)	Off-chain (Data availability layer, no guarantee that each user holds data by themselves)
Sequencing	Two persons concerned	An operator for each Plasma chain	Sequencers	Sequencers	Multiple “operator”s
Security solution	Monitoring Service (watchtower)	Fraud Proof Crypto- economic incentives incl staking bond / rewards) Mass Exit	Fraud Proof Crypto- economic incentives incl staking bond / rewards	Validity Proof	Validity Proof
Security challenge	・Is monitoring service secure enough?	・How are operators chosen? ・PoC > CoC ・Delays in transaction finality ・Challenges of mass exit ・How can double spending across chains be prevented?	・How are sequencers chosen? ・PoC > CoC? ・Delays in transaction finality	・How are sequencers chosen? ・Special hardware is needed for validity proof - centralization ・Slower subjective finality?	・How are sequencers chosen? ・Special hardware is needed for validity proof - centralization ・Is there any safety measure for block withholding attack? ・Slower subjective finality?

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1. State Channel

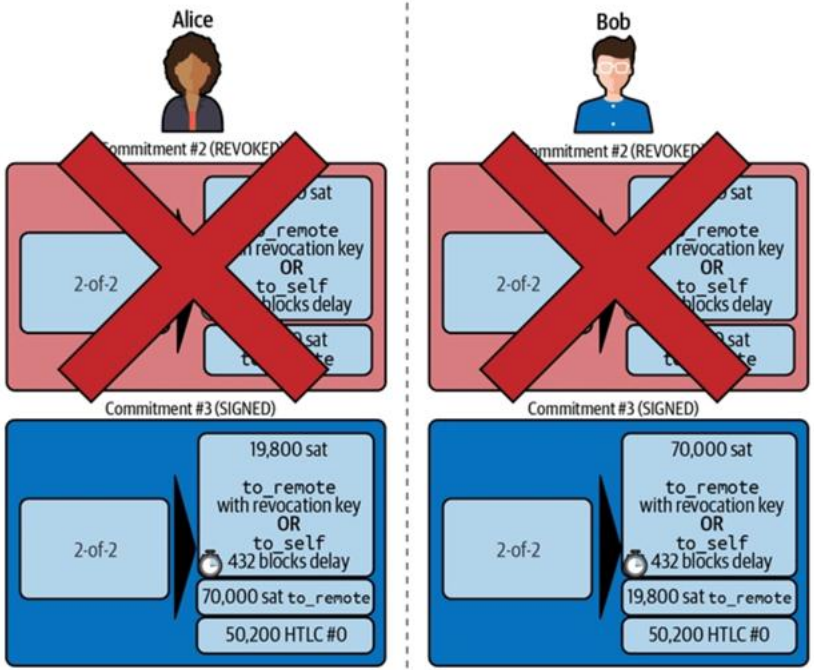
State Channel

Data	Sequencing	Proof type
Local environments of 2 parties concerned (Off-chain)	2 persons concerned	Both parties sign the “Exit Transaction”

- To open a Payment Channel between two parties, a certain amount of funds is deposited into a Multi-Sig Address (or Smart Contract) that requires both parties' signatures. Once the Payment Channel is established, multiple transactions can take place between the two parties (with the possibility of intermediating nodes). After each transaction, both parties sign the latest state, reflecting the updated balance or conditions.

- If either party wishes to close the Payment Channel, they submit the transaction reflecting the latest state, with both parties' signatures, to the main chain (Ethereum). Once this transaction is approved on the main chain, the funds are distributed according to the final state. If one party is offline or uncooperative, the other party can generate an Exit transaction independently, setting a challenge period for the offline party to respond.

- To prevent fraudulent Exits when one party is offline, a third-party service known as a monitoring service (Watchtower) is available. Both parties can share the hash of each state and a Penalty Transaction with the Watchtower after every transaction. The Watchtower monitors the network for any unauthorized Exit attempts and can act to prevent them if necessary.



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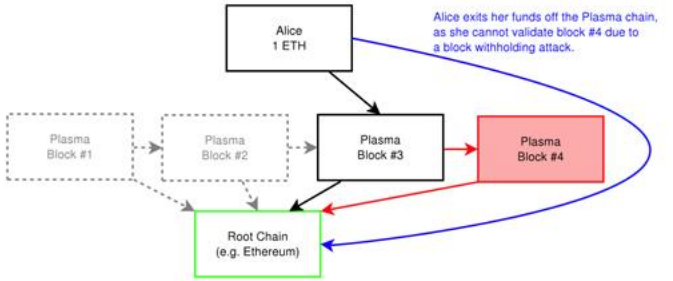
2. Plasma

Plasma

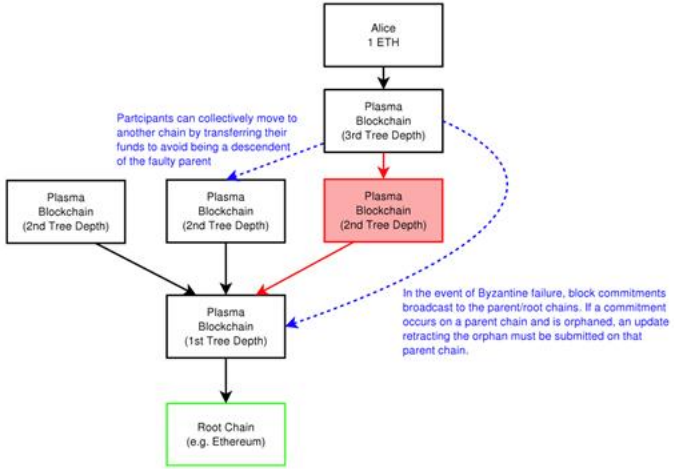
Data	Sequencing	Proof type
Depends on operators (off-chain)	Operators	Fraud Proof

- In a Plasma chain, if an Operator withholds the block information they generated and does not disclose it to network participants, the state transitions of the entire chain become unclear, creating a situation where fraudulent transactions could be possible (this is known as a block withholding attack).
- When fraudulent behavior is detected, proving whether the User or the Operator is correct requires revealing the current state on-chain. However, if the data is being concealed, proving this becomes difficult. Consequently, users are incentivized to exit that Plasma chain quickly and move to another Plasma chain.
- In the event of a block withholding attack, users may collaborate with other participants to plan a Mass Exit. The Exit Processor verifies the validity of the data on the Plasma chain, collects signatures from all participants, and then broadcasts the exit request to the target Plasma chain with a challenge period.
- The Exit Processor validates the data using block data from before the block withholding attack, information provided by users, and data recorded on Ethereum. Mechanisms such as bond deposits and the use of multi-signatures are introduced to prevent fraud.
- If a challenge is submitted regarding transactions after the block withholding attack, another participant can challenge the challenger by proving that the challenger cannot provide valid payment proof. This process helps prevent the forfeiture of bonds due to an unintentional and "inaccurate" Mass Exit.

Mass Exit



Source



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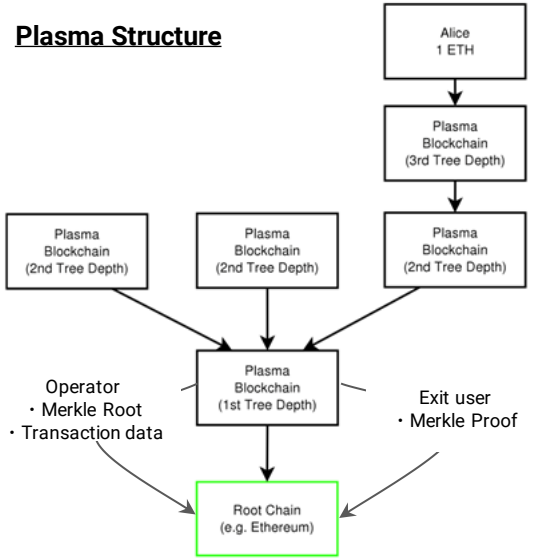
2. Plasma

Plasma

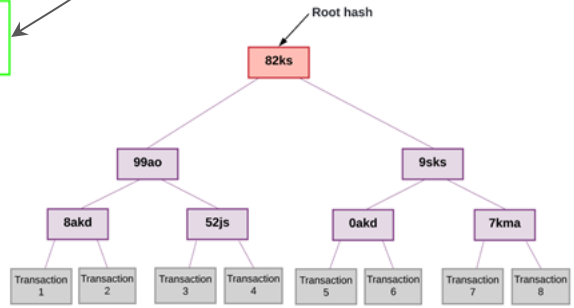
Data	Sequencing	Proof type
Depends on operators (off-chain)	Operators	Fraud Proof

- Multiple Plasma chains (which are not strictly blockchains) operate independently from Ethereum, and the Merkle Roots of transaction blocks on each Plasma chain (a unique hash value identifying all transactions within a block) are periodically recorded onto a smart contract on Ethereum.
- Plasma chains are structured hierarchically, with Parent chains managing multiple Child chains. The Merkle Root of a Child chain is first recorded in the block of its Parent chain and ultimately recorded on the Ethereum smart contract via the Parent chain.
- Each Plasma chain is managed by a different group of Operators who are responsible for transaction validation, block generation, and recording the Merkle Root on the Parent chain or directly on Ethereum. The transaction data is maintained by the Operators, who are also obligated to store it off-chain in an appropriate public location.
- Fraud Proof: If an Operator or a user attempting to exit commits fraud, other users can challenge the Merkle Root or Merkle proof recorded on the Ethereum smart contract within a specified period. If the challenge is successful, the fraudster's pre-deposited ETH/ERC-20 tokens will be slashed as a penalty.

Plasma Structure



Merkle Tree



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3. Rollup

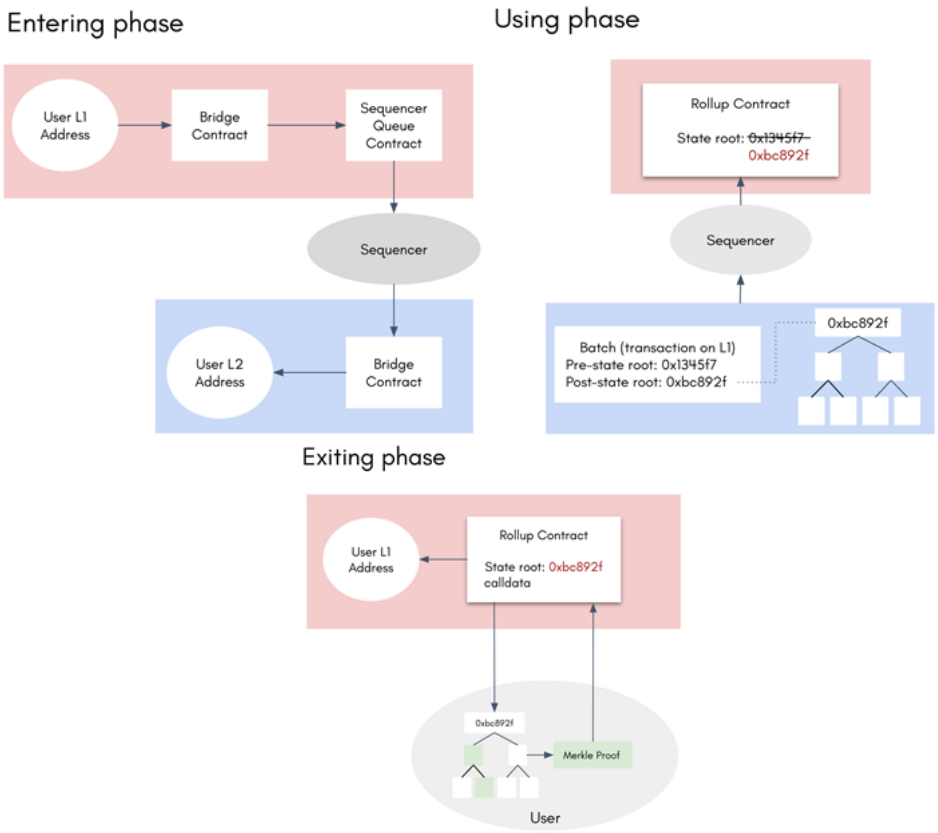
Rollup

- First, users of Rollup need to deposit funds into a Bridge Contract on Ethereum. After the deposit, a Sequencer mints an equivalent amount of tokens on the Layer 2 (L2) by interacting with the Bridge Contract on the L2 after the deposit is registered in the Sequencer Queue Contract on Ethereum.

- The Sequencer is responsible for verifying each transaction on the L2, maintaining the state tree of all executed transactions, and periodically creating a state root. The Sequencer then records and updates this state root along with the compressed data of all transactions on Ethereum as calldata.

- When a user deposits tokens into the Rollup Contract on the L2 and initiates an Exit Request, they generate a Merkle Proof using the state root stored as calldata on Ethereum by the Sequencer, along with the necessary transaction data. This Merkle Proof (a cryptographic proof that their transaction is included in the Merkle Tree) is then submitted to the Rollup Contract on Ethereum. If the Rollup Contract validates the Merkle Proof, the user can withdraw their funds from the Rollup Contract.

- To ensure censorship resistance of the data recorded by the Sequencer and users in the Rollup Contract on Ethereum, Optimistic Rollup and ZK Rollup mechanisms are frequently employed.



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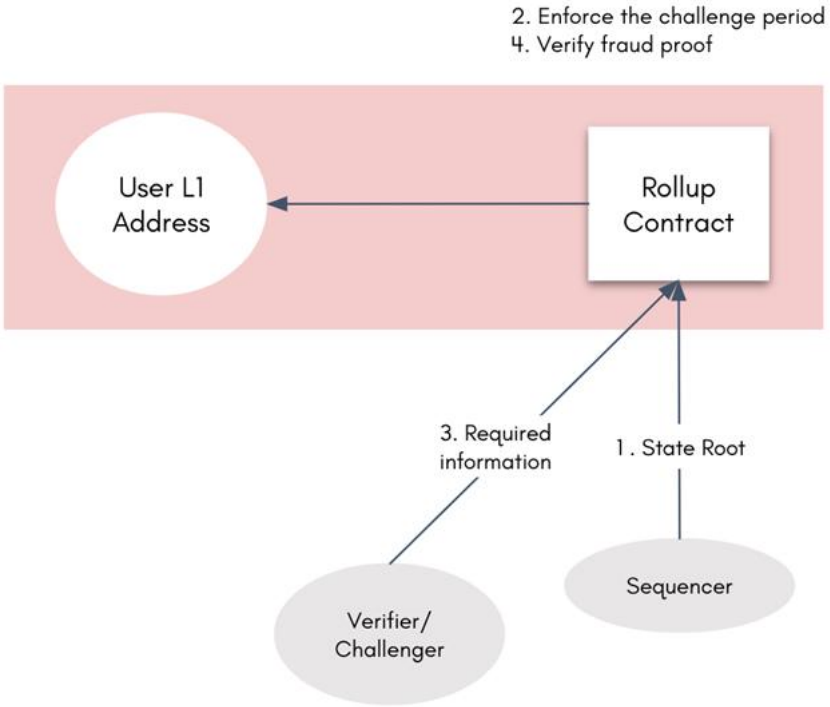
3-1. Optimistic Rollup

Optimistic Rollup

Data	Sequencing	Proof type
Calldata (on-chain) or EIP-4844 (off-chain)	Sequencers	Fraud Proof

- After the user submits a Merkle Proof to the Rollup Contract on Ethereum, the transaction's finality is not confirmed until the challenge period has passed. In the case of Optimistic Rollup, this challenge period typically lasts around 1 to 2 weeks.
- During the challenge period, a Verifier/Challenger can dispute the validity of the data provided by the user or the Sequencer. If they believe the data is incorrect, they can submit a challenge to the Rollup Contract on Ethereum. The information that should be provided when submitting a challenge includes the following:
  - Dispute state transition
  - The pre-state root
  - Rollup state data
  - Their version of the post-state root
- When a challenge is submitted, the Rollup Contract on Ethereum recalculates the post-state root using the information provided by the challenger. If the recalculated post-state root matches the one provided by the challenger, the user's exit is deemed invalid.
- If the challenge is successful, the funds pre-deposited by the Verifier/Challenger or the user are subject to slashing, which incentivizes correct behavior.



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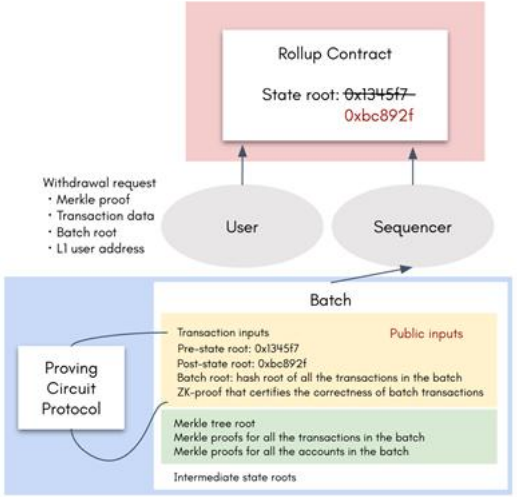
3-2. ZK Rollup

ZK Rollup

Data	Sequencing	Proof type
Calldata (on-chain) or EIP-4844 (off-chain)	Sequencers	Validity Proof

- Since the Sequencer regularly submits a Validity Proof along with the State Root to the Rollup Contract on Ethereum, the Rollup Contract can immediately verify on-chain that all submitted data is correct. This eliminates the need for a challenge period when a user wants to exit.
- The Sequencer validates each transaction, batches them together, and uses a Proving Circuit Protocol to generate a ZK-Proof (Validity Proof) that proves the correctness of all transactions within the batch. The computation of the Validity Proof involves all transaction data, the Merkle Root, data from state transitions, the old and new State Roots, and the signatures and public keys required to verify the validity of each transaction.
- The Sequencer submits the latest State Root along with the Validity Proof to the Rollup Contract on Ethereum. Only after the Validity Proof is verified can the State Root in the Rollup Contract on Ethereum be updated.
- If a user wishes to exit, they can submit an Exit Request to the Rollup Contract on Ethereum, allowing them to withdraw their funds immediately without any delay.




```
zkProof = GenerateProof(
    oldStateRoot,
    newStateRoot,
    transaction,
    validStateTransition
)

isValid = VerifyProof(
    zkProof,
    oldStateRoot,
    newStateRoot
)
```

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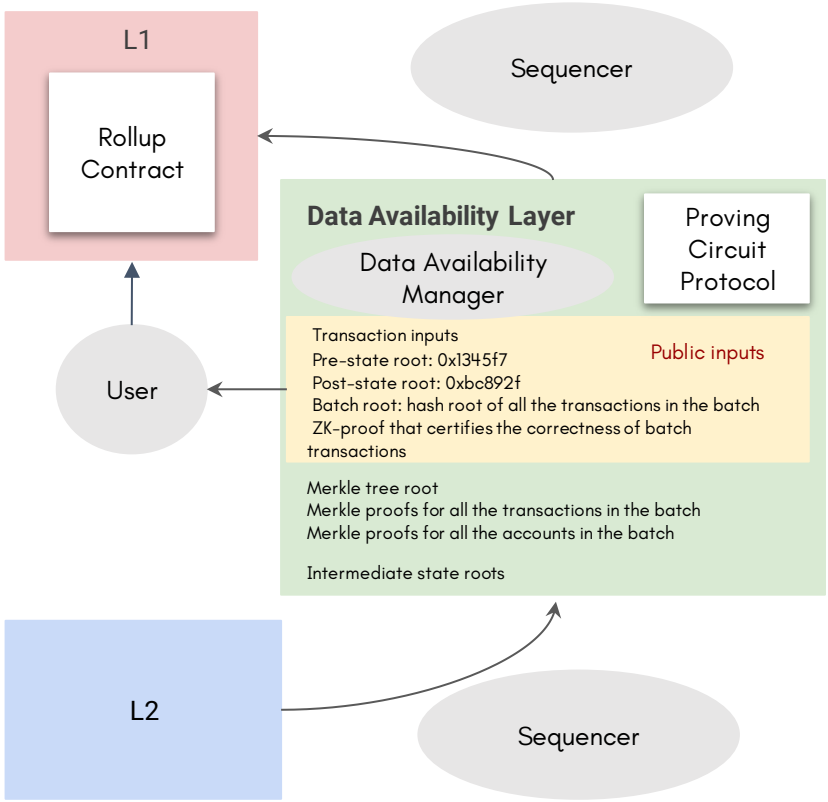
4. Validium

Validium

Data	Sequencing	Proof type
Data Availability Layer (off-chain)	Data Availability Committee / Sequencer	Validity Proof

- Validium, an evolution of ZK-Rollup, addresses similar scalability challenges but with a key difference in how transaction data is handled. In ZK-Rollup, all transaction data is compressed and stored on-chain as calldata. In contrast, Validium stores this data off-chain in a Data Availability Layer.
- The Data Availability Layer is managed by a Data Availability Committee, which consists of multiple Data Availability Managers (DAMs) or Sequencers. These Sequencers are responsible for validating transactions on Layer 2 (L2) and submitting and updating the State Root on Ethereum with a Validity Proof, similar to how it works in ZK-Rollup.
- Since all transactions are stored off-chain, incentive mechanisms are in place to ensure DAMs act correctly, such as slashing pre-deposited tokens if they behave maliciously. However, like Plasma, Validium is also susceptible to block withholding attacks, where DAMs or Sequencers may withhold block information, potentially leading to fraudulent activity.



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Data Challenges

	Plasma	Optimistic Rollup	ZK-Rollup	Validium
Data storage	Depends on operators (off-chain)	Calldata (on-chain) or EIP-4844 (off-chain)		Data Availability Layer (off-chain)
Challenge	• Plasma chains often have a limited number of Operators. • In the event of a block withholding attack, there may be no original data available to generate a Fraud Proof, leaving Mass Exit as the only viable solution. • Mass Exit requires substantial computational resources and gas fees for each transaction and its validity proof, raising concerns about whether all participants would be adequately motivated to opt for Mass Exit. • While cross-chain double-spending on Plasma Chains is theoretically disincentivized through challenges via Fraud Proofs, it is uncertain whether this mechanism alone is sufficient to detect and prevent all instances of double-spending in practice.	• When using calldata, there is a gas cost associated with writing all compressed Layer 2 transactions on-chain. • While the use of calldata or EIP-4844 reduces the likelihood of block withholding attacks, if a Data Availability Layer is utilized, similar challenges to those found in Validium may arise.		• Since all data is managed off-chain and the reliability of the data depends entirely on the Data Availability Committee (DAC), if the Data Availability Managers or Sequencers within the DAC are not sufficiently decentralized, it becomes easier for them to collude and carry out block withholding attacks. • If the Data Availability Managers are centralized, there are risks associated with technical failures of the DAMs, as well as the potential for them to censor specific users or transactions. • In the event that the DAC fails to function properly, there may be an "escape hatch" mechanism that allows users to recover their assets (further investigation required).

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1. Basics - Calldata

Calldata

Cost	Scalability	Security
△	△	◎

- Calldata refers to the bytes sent as part of a transaction that are permanently recorded on the blockchain. It has been the most cost-effective method for permanently storing data on the blockchain, which is why it has been used in rollups to store the transaction data necessary for submitting Fraud Proofs. However, the amount of data stored in calldata directly impacts gas fees.
- To improve the efficiency of the main chain, alternative off-chain data storage methods are being explored, such as through EIP-4844 and the use of Data Availability Layers.

```
nonce: 0, // アカウントのnonce
gasPrice: ethers.utils.parseUnits('20', 'gwei'), // ガス価格（Gwei単位）
gasLimit: ethers.utils.hexlifv(100000), // ガスリミット
to: '0xContractAddress', // スマートコントラクトのアドレス
value: ethers.utils.parseEther('0.1'), // 送金するETHの量（ether単位）
data: '0x', // スマートコントラクトへの関数呼び出しや追加データ（通常はABIエンコードされた関数呼び出し）
chainId: 1 // チェーンID（1はEthereumメインネット）
```

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1. Basics - EIP-4844

EIP-4844

Cost	Scalability	Security
○	○	◎

- A "Blob" is a new data format introduced by EIP-4844, designed to be handled differently from traditional transactions on Ethereum. Unlike standard transactions, Blobs are not stored directly on-chain; instead, they are transmitted and stored off-chain.
- The posting of Blobs comes with a cost, set at 1 wei (10^18 ETH) per byte, which is minimal compared to the gas fees required for executing transactions.
- Blobs are increasingly being used in Optimistic Rollups, where it's necessary to publish Layer 2 transaction data to enable the submission of Fraud Proofs. In the context of Optimistic Rollups, because there is a mechanism for Proofs, there is no need to permanently store all Layer 2 transaction data on-chain solely for verification purposes. The use of Blobs helps optimize this process, ensuring that the necessary data is accessible for verification without incurring the higher costs of on-chain storage.

https://blobscan.com/address/0x1c479675ad559DC151F6Ec7ed3FbF8ceE79582B6

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1. Basics - Data Availability Layer

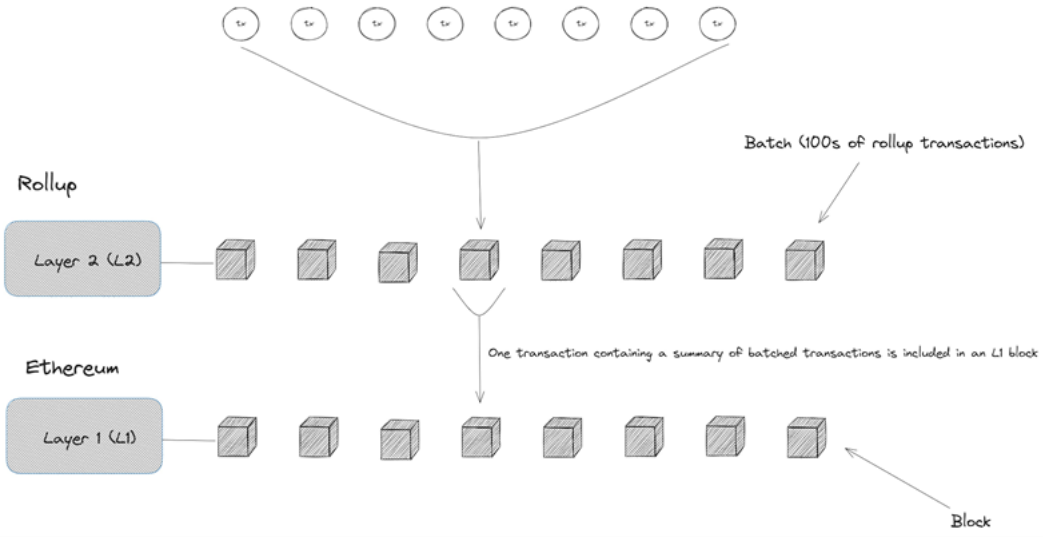
Data Availability Layer

Cost	Scalability	Security
◎	◎	○ (?)

- In decentralized protocols, the off-chain storage of transaction data on Layer 2 is decentralized to prevent the Sequencer (Operator) from compromising data availability through malicious actions. To address this, a Data Availability Committee (DAC) is used, consisting of a group of trusted "Managers" responsible for managing the storage and availability of Layer 2 transaction data across multiple locations.

- In many DAC systems, there is mutual monitoring among the "Managers," and incentive mechanisms are in place where any misconduct leads to the slashing of staked assets. This setup encourages proper behavior among the Managers.

- However, solutions to the risk of collusion among Managers remain limited. Current approaches include increasing the number of data custodians (enhancing decentralization) and introducing external monitoring mechanisms by third parties. These solutions are still being explored and developed to strengthen the system's resilience against collusion.



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Sequencing Challenges

	State Channel	Plasma	Optimistic Rollup	ZK-Rollup	Validium
Interoperation	Monitoring Service (Watchtower)	Operators	Sequencers	Sequencers	Data Availability Committee / Sequencer
Challenges	• Monitoring Services might not be used in sufficient numbers, increasing vulnerability to attacks or monitoring failures. • Multisig Address and other key management issues could lead to security risks due to potential mismanagement or key compromise.	• Plasma chains often have a small number of Operators, which may limit risk diversification. • Vulnerabilities in smart contracts on both Ethereum and Plasma chains. • Is the bond setup consistently ensuring that Proof of Commitment (PoC) outweighs Collusion of Commitment (CoC)?	• There is a possibility that the Sequencers are too few, leading to insufficient decentralization. • Vulnerabilities in smart contracts on Ethereum. • Is the bond setup consistently ensuring that Proof of Commitment (PoC) outweighs Collusion of Commitment (CoC)?	• There is a possibility that the Sequencers are too few, leading to insufficient decentralization. • Vulnerabilities in smart contracts on Ethereum. • Is the bond setup consistently ensuring that Proof of Commitment (PoC) outweighs Collusion of Commitment (CoC)?	• There is a possibility that the DAMs and Sequencers are too few, leading to insufficient decentralization. • Vulnerabilities in smart contracts on Ethereum. • Is the bond setup consistently ensuring that Proof of Commitment (PoC) outweighs Collusion of Commitment (CoC)?

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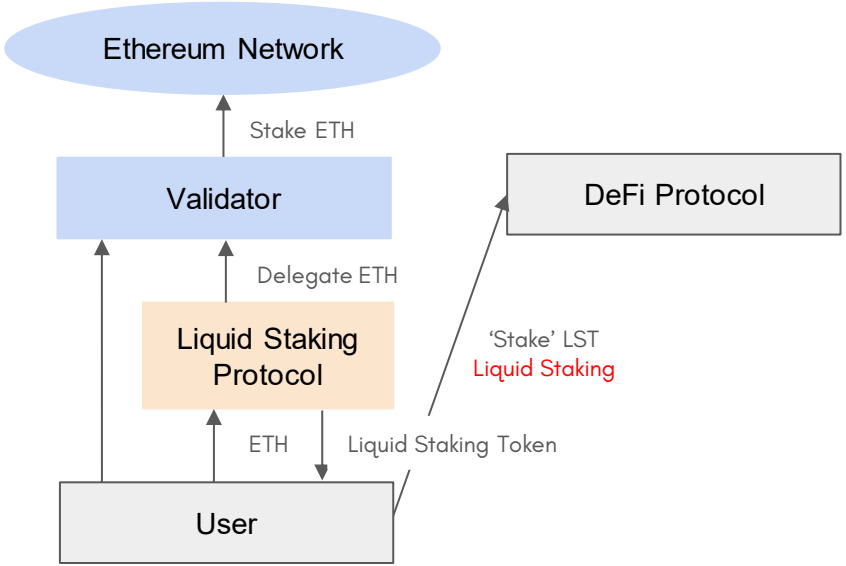
Liquid Staking

How does it work?



Players

- Lido stETH
- Rocket rETH
- Ether.fi eETH

Description

- When users provide ETH into a liquid staking protocol such as Lido and Rocket Protocol, they receive derivative tokens that represent their staked assets. For example, if a user stakes ETH, they receive stETH in return.
- These derivative tokens (Liqui Staking Tokens) are liquid and can be traded on secondary markets, used as collateral for loans, or employed in other DeFi protocols such as yield farming or liquidity mining.
- At any time, users can redeem their LSTs for ETH along with any accrued staking rewards. This redemption process is typically managed through smart contracts to ensure transparency and security.

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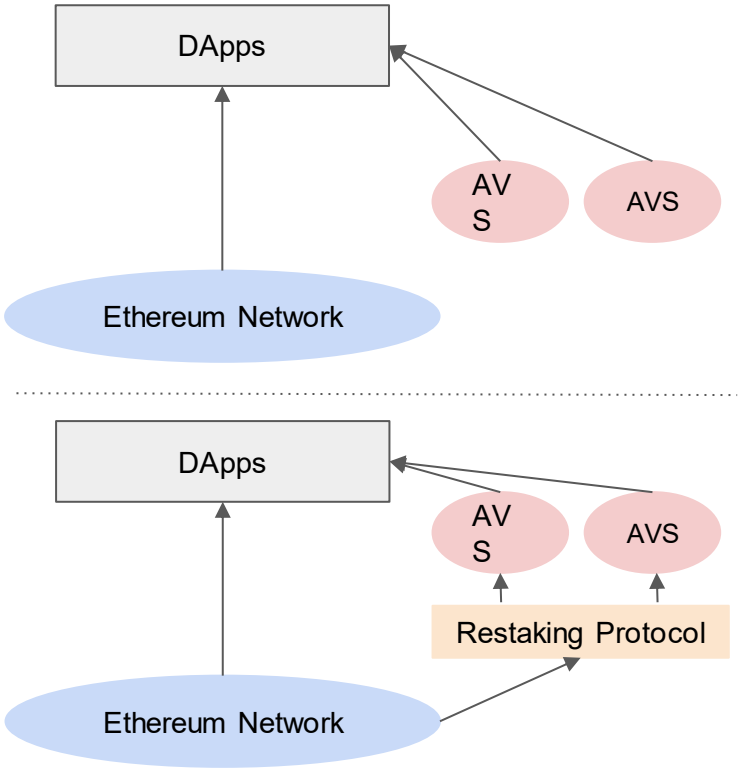
What is restaking for?



Actively Validated Service (AVS)

- Actively validated services, including layer 2 solutions, cross-chain interoperability and oracle services, are services provided by validators that go beyond basic transaction validation and block production.
- AVSs often issue new tokens and create a security pool that ensures validators have a stake in the proper functioning of the AVS (smart contract). Validators who misbehave or fail to perform their duties can be penalized, which aligns their incentives with the overall health of the network.

The value of restaking

- EigenLayer allows existing validators who are already staking on Ethereum to participate in securing AVSs. This means that rather than having to attract and incentivize new validators, EigenLayer can utilize the security and trust of an already established validator network.
- If a restaker is found to be engaging in malicious behavior, the network will trigger slashing through restaked ETH.

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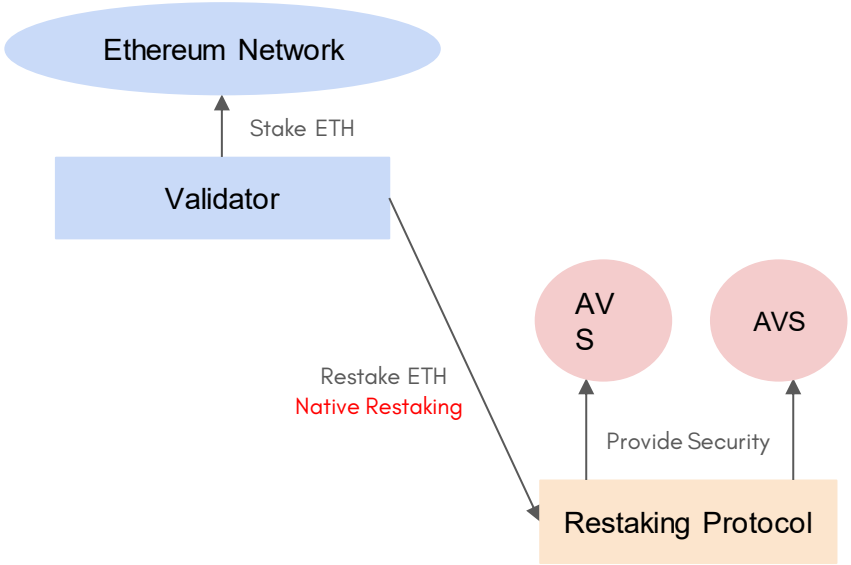
Native Restaking

How does it work?



Restaking Protocol

- EigenLayer

Native Restaking

- As explained in the Staking section, validators stake their tokens in Ethereum Network, and their staked ETH are locked into the primary Ethereum staking contract.
- To enable restaking through EigenLayer, validators need to set their withdrawal credentials to an EigenPod, which is a smart contract that manages their staked ETH for restaking purposes.
- Validators (restakers) register with the EigenLayer protocol, which involves interacting with the EigenLayer smart contract. Validators can select the actively validated services (AVSs), including layer 2 solutions, cross-chain bridges, data availability layers, oracles, etc, which they want to support.
- Validators (restakers) need to set up dedicated nodes to monitor transactions and events on selected AVS's smart contracts, enabling real-time tracking of the smart contract behavior and ensuring proper validation.
- If validators (restakers) get slashed through EigenLayer, Eigenpod is responsible for seizing a portion of the staked ETH from them according to the slashing rules and penalties defined in the AVS.

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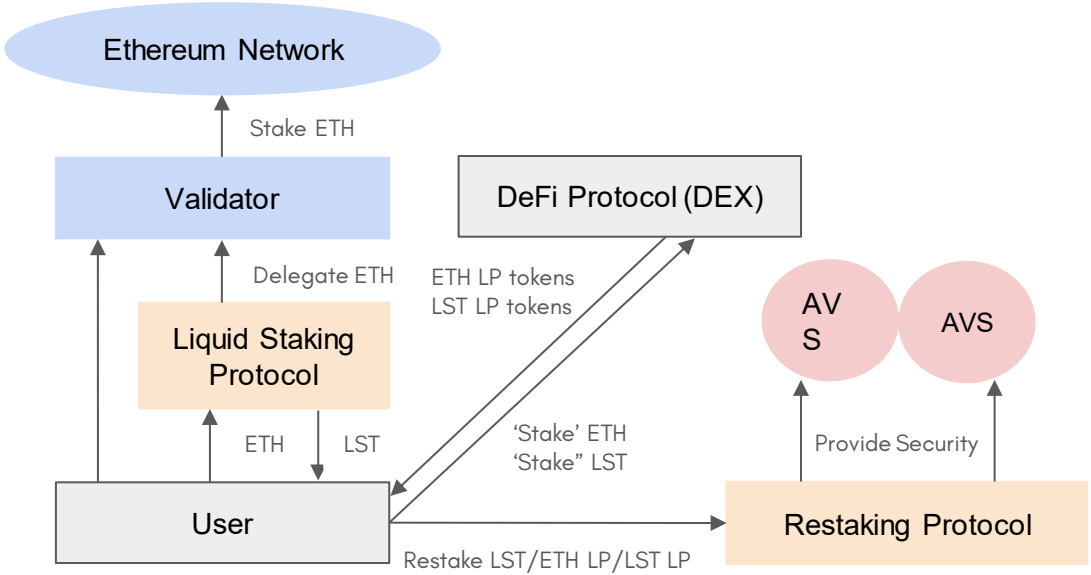
Other Restaking

How does it work?



LST Restaking / ETH LP Restaking / LST LP Restaking

LST Restaking

- Validators restake by staking their LSTs, ETH already restaked via liquid staking protocols like Lido and Rocket Pool, by transferring their LSDs into the restaking protocol smart contracts

ETH LP Restaking

- Validators stake the Liquidity Provisioning (LP) token of a pair which includes ETH. An LP token represents a share of the total liquidity contained in a DeFi exchange such as Uniswap and Curve.

LST LP Restaking

- Validators stake the LP token of a pair which includes a liquid staking ETH token, such as Curve's stETH-ETH LP token,

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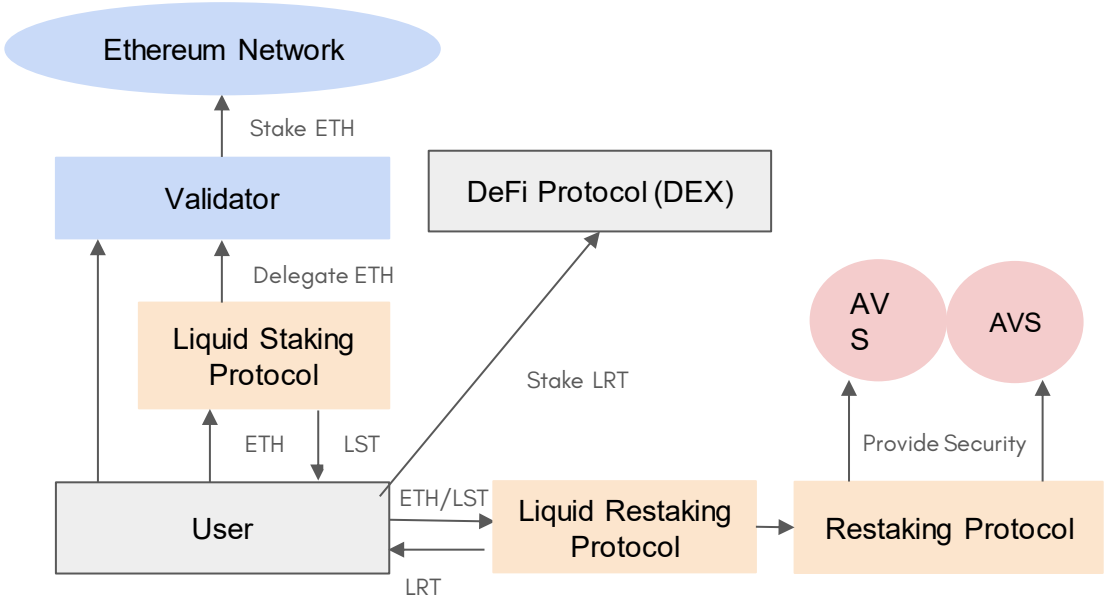
Liquid Restaking

How does it work?



Liquid Restaking Protocol

- Ether.fi
- Puffer
- Renzo

Liquid Restaking

- Instead of directly restaking LST in a restaking protocol, users deposit LST into a liquid restaking protocol.
- In return, they receive liquid restaking tokens (LRT) from the liquid restaking protocol.
- Users can then use the LRT obtained from the liquid restaking protocol in DeFi protocols (such as providing liquidity) to earn additional rewards.

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Analysis of factors that could lead to vulnerabilities in financial stability

	Fungibility	Reward	Trust	Redemption
Native Staking	No	• Receive rewards in exchange for the transaction/block validation & block generation • ETH denominated rewards	• Yourself for properly dealing with Ethereum validator work • Security of Ethereum	• Limited by Ethereum
Pooled Staking	No	• Receive a part of rewards from delegated validators' work • ETH denominated rewards	• Delegated Validators for properly dealing with Ethereum validator work • Security of Ethereum	• Limited by delegated validators, fundamentally by Ethereum
Liquid Staking	Yes if LSP adopts rebasing token model / No if LSP adopts reward-bearing token model	• Receive a part of rewards from LSP delegated validators' work • Receive additional rewards from DeFis you support with LST • LST denominated rewards	• LSP delegated validators for properly dealing with Ethereum validator work • Security of LSP and Ethereum	• Limited by LSP, fundamentally by Ethereum
Native Restaking	No	• Receive rewards in exchange for validator work to Ethereum and AVSs that you support • ETH and/or AVS tokens denominated rewards	• Yourself for properly dealing with validator work to Ethereum and AVSs • Security of Ethereum and AVSs	• Limited by Ethereum and AVSs that you support
LST Restaking	No *assume that LSPs which issue the LSTs adopts rebasing token model	• Receive a part of rewards from LSP delegated validators' work to Ethereum and AVSs • LST denominated rewards	• LSP delegated validators for properly dealing with validator work to Ethereum and AVSs • Security of LSP, Ethereum and AVSs	• Limited by LSP, fundamentally by Ethereum and AVSs LSP delegated validators support
Liquid Restaking	Yes if LRP adopts rebasing token model / No if LRP adopts reward-bearing token model *assume that LRPs accept only ETH	• Receive a part of rewards from LRP delegated validators' work to Ethereum and AVSs • Receive additional rewards from DeFis you support with LRT • LRT denominated rewards	• LRT delegated validators for properly dealing with validator work to Ethereum and AVSs • Security of LRP, Ethereum and AVSs	• Limited by LRP, fundamentally by Ethereum and AVSs LRP delegated validators support

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Vulnerability vs Shock

Vulnerability

- Vulnerability is an underlying 'state' within a decentralized financial system that has the potential to amplify that 'state' or trigger a new 'state' in the event of a shock or stress event."

Shock

- A shock is a sudden disturbance within the decentralized financial system that triggers an amplification of vulnerabilities and/or leads to new vulnerabilities emerging.

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Taxonomy of Vulnerability

	L1 (Ethereum)	LSP	RP	LRP	AVS
Asset Price	• ETH volatility • ETH mispricing	• Depeg of LST		• Depeg of LRT	• AVS token volatility • AVS token mispricing
Asset Quality	• A centralization of / a decrease in #validators • A decrease in #users • Low staking rewards	• A bad LSP DV selection • A centralization of / a decrease in #LSP DVs • A decrease in #users • Low rewards given to users	• Restake to high-risk AVSs • A centralization of / a decrease in #restakers in AVS • A decrease in #users	• Restake to high-risk AVSs • A bad LRP DV selection • A centralization of / a decrease in #LRP DVs • Low rewards given to users	• A decrease in #users • Low restaking rewards
Funding/ Liquidity	• Limited redemption of staked ETH	• A high proportion of ETH deposited in LSP • Limited redemption • Non-fungibility of LST	• A high proportion of ETH restaked via RP • Limited redemption	• A high proportion of ETH or LST deposited in LRP • Limited redemption • Non-fungibility of LRT	• Limited redemption of staked AVS token
Governance	• Slow incident response • Centralized operation	• Slow incident response • Centralized operation • Transparency in LSP DV • Economic security	• Slow incident response • Centralized operation • Economic security	• Slow incident response • Centralized operation • Transparency in LRP DV • Economic security	• Slow incident response • Centralized operation • Economic security
Security	• Cryptographic security	• Technical bugs • Vulnerable smart contract	• Technical bugs • Vulnerable smart contract	• Technical bugs • Vulerable smart contract	• Technical bugs • Vulnerable smart contract
Information	• Regulatory uncertainty • Speculative information	• Temporary hype driven by users' reward expectations • Regulatory uncertainty • Speculative information	• Temporary hype driven by users' reward expectations • Regulatory uncertainty • Speculative information	• Temporary hype driven by users' reward expectations • Regulatory uncertainty • Speculative information	• Temporary hype driven by users' reward expectations • Regulatory uncertainty • Speculative information
Linkage	• Cross-chain protocols • Other chains	• Stablecoin issuers • Lending protocols • DEXs • Insurance issuers • Oracles • Cross-chain protocols • Other chains	• Insurance issuers • Oracles • Cross-chain protocols • Other chains	• Stablecoin issuers • Lending protocols • DEXs • Insurance issuers • Oracles • Cross-chain protocols • Other chains	• Protocols that use the AVSs (Layer2, DA, Oracle, etc) • Insurance issuers • Oracles • Cross-chain protocols • Other chains

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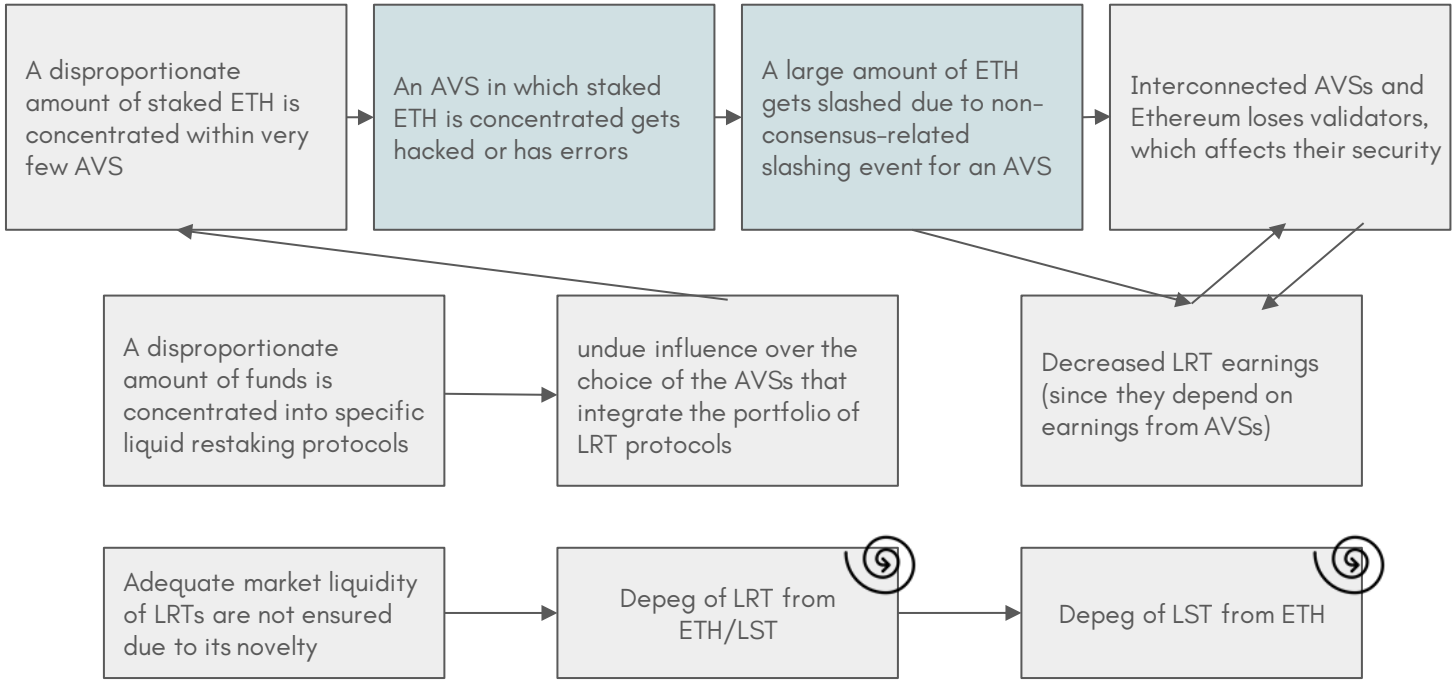
Scenario Analysis: Liquid Restaking



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Layer2 meetup @ SF

Blockchain Governance Initiative Network

14:45 - 15:00 Coffee Break

sponsored by DGUS

Layer2 meetup @ SF

Blockchain Governance Initiative Network

15:00 - 16:30 Workshop Session 2

Wallet Governance

sponsored by DGUS

WORKSHOP RULE

Blockchain Governance Initiative Network

CHATHAM HOUSE RULE

When a meeting, or part thereof, is held under the Chatham House Rule, participants are free to use the information received, but neither the identity nor the affiliation of the speaker(s), nor that of any other participant, may be revealed.

All workshop sessions in Layer2 Meetup will be held in Chatham House Rule.

WORKSHOP RECORDING USAGE

The recording of all workshop sessions will not be distributed and will not be used internally to create anonymized reports, which will be published on BGIN's website.

Layer2 meetup @ SF

Blockchain Governance Initiative Network

16:30 - 16:45 Coffee Break

sponsored by DGUS

Layer2 meetup @ SF

Blockchain Governance Initiative Network

16:45 - 18:15 Workshop Session 3

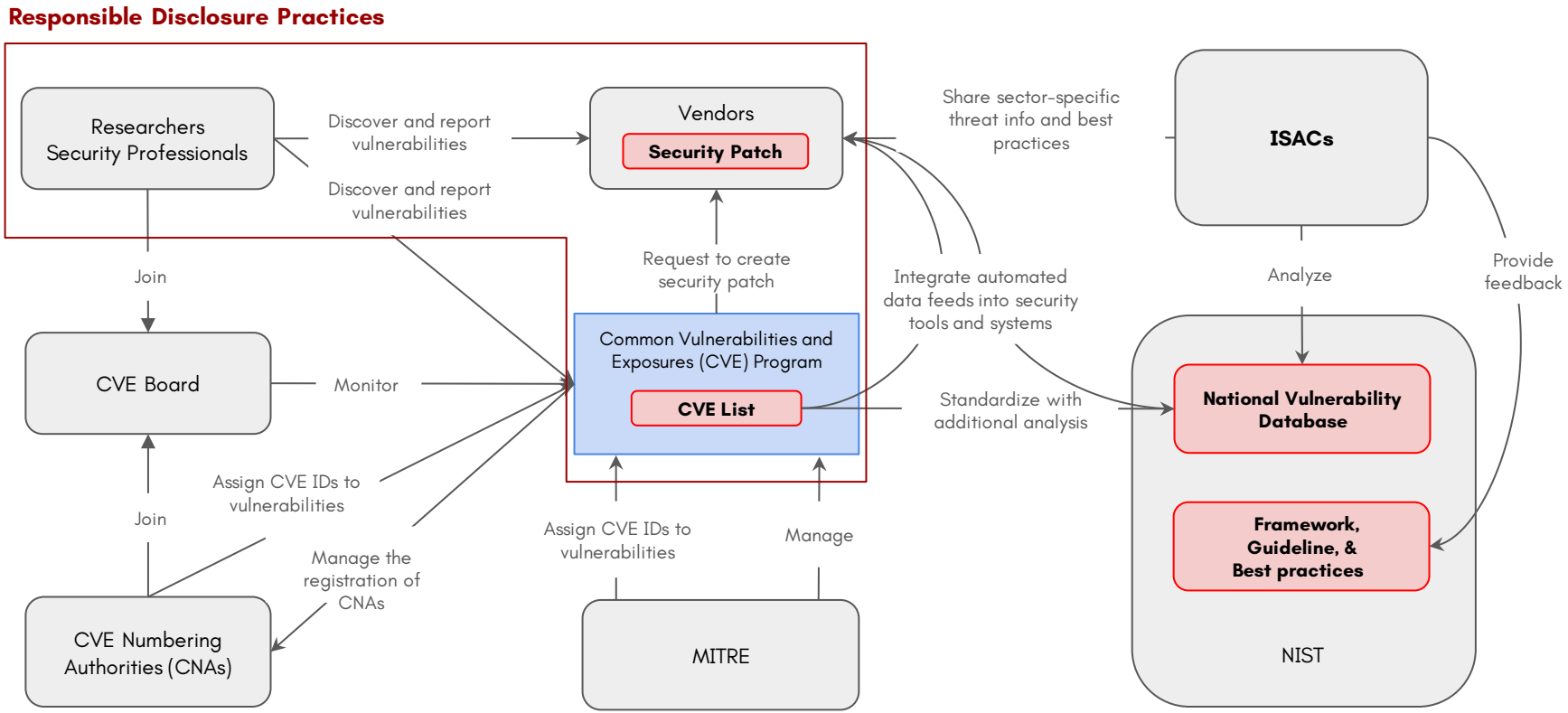
Cybersecurity Standards

sponsored by DGUS

BGIN's Potential Contribution to Cybersecurity Information Sharing Alignment in the Crypto Industry

Blockchain Governance Initiative Network (BGIN)

General Cybersecurity Info Sharing Alignment in the U.S



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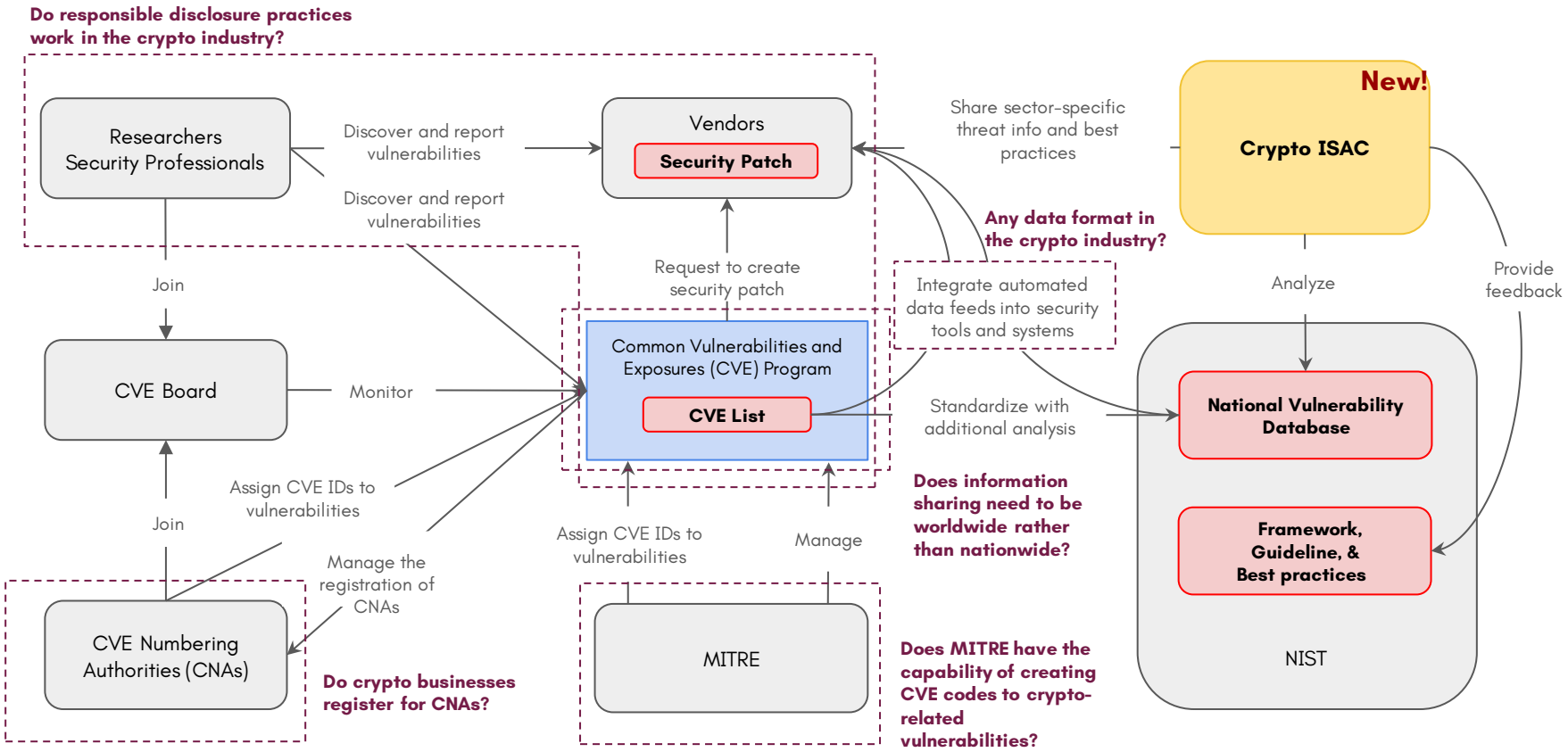
Can the alignment work in the crypto industry?



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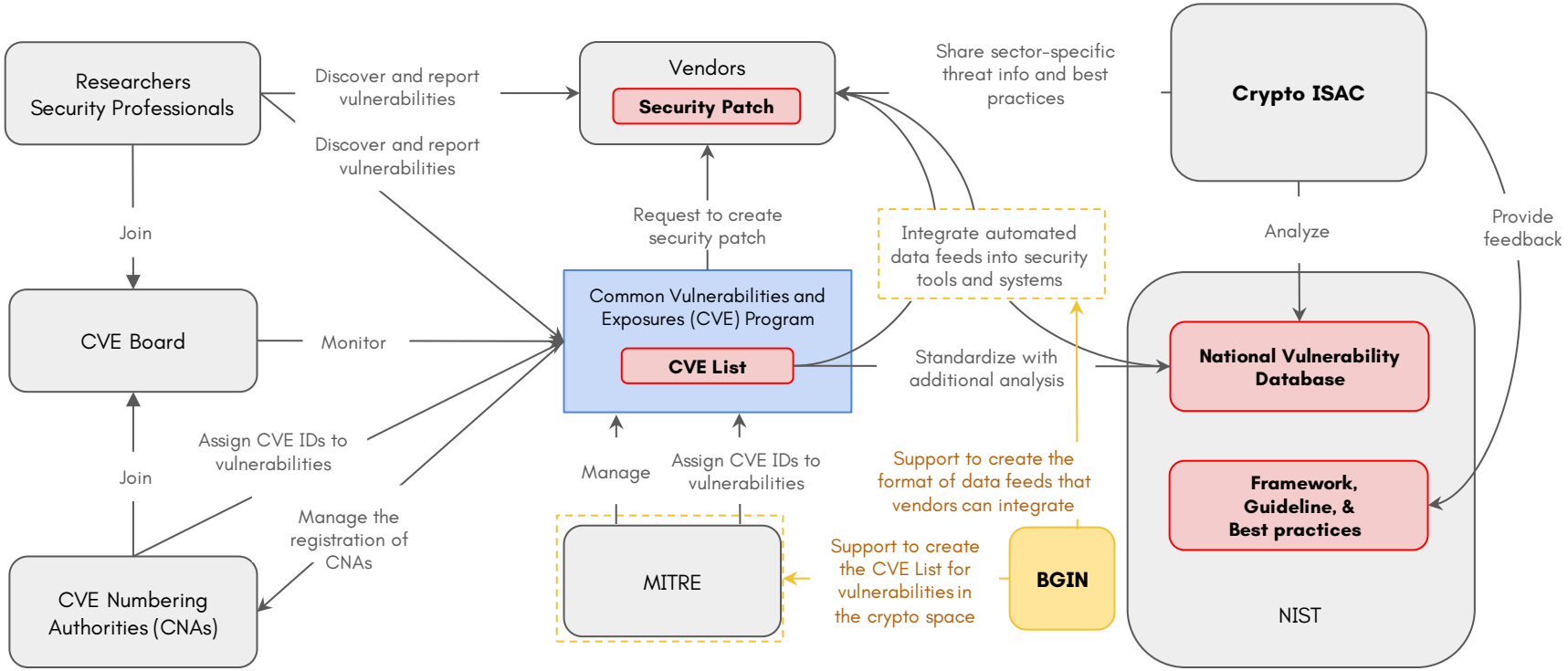
What can BGIN potentially do?



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